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RESEARCH ARTICLE

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Assessing dual hair sampling for isotopic studies of grizzly bears

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Rationale: The stable isotope ratios of carbon (δ^{13}C values), nitrogen (δ^{15}N values) and sulfur (δ^{34}S values) in bear hair can be used to obtain information on dietary history. Sample protocols often require hair sampling from multiple anatomical locations; however, there remains a question as to whether this is necessary for isotopic studies of hair. The purpose of this study was to determine whether significant differences can be observed for the δ^{13}C, δ^{15}N and δ^{34}S values between paired hair samples taken from the rump and shoulder of grizzly bears (*Ursus arctos*).

Methods: Paired hair samples were collected from the rump and the shoulder of 81 grizzly bears in the Yukon, Canada. Hair samples were analyzed using a thermal combustion elemental analyzer coupled with a continuous flow isotope ratio mass spectrometer.

Results: Statistical comparisons of paired hair samples for both males and females showed no meaningful differences in δ^{13}C, δ^{15}N and δ^{34}S values in hair taken from the rump and shoulder, and any observed differences fell within the instrumental error.

Conclusions: Based on these results, hair may be safely sampled on either the rump or the shoulder without loss of isotopic information and thus this finding allows for refinement of sampling.

1 | INTRODUCTION

The dietary history and food habits of bears are often studied by stomach content analysis[1] or scat analyses.[2,3] Limitations to these approaches have been noted[4] and stable isotopic analyses of bear tissues are now increasingly being applied to diet and behavioural studies of bear populations.[5-8] Hair is particularly useful for analyses as it does not undergo tissue remodelling and large sample sizes can be collected in a relatively non-invasive manner compared with other tissues such as blood. It further preserves temporal dietary information as hair forms incrementally over time in a cyclic pattern.[9,10] Thus, the isotopic ratios of carbon (δ^{13}C values), nitrogen (δ^{15}N values) and sulfur (δ^{34}S values) in bear hair can give an indication of food and water sources.[6,11]

δ^{13}C values provide information on the relative contribution of C_3 and C_4 plants consumed, where higher consumption of C_4 plants, or indirect consumption of human foods, results in higher δ^{13}C values.[12] δ^{15}N values provide information pertaining to primarily protein consumption and secondarily trophic level, where a higher δ^{15}N value signifies feeding of a marine or meat-based diet.[7,13,14] δ^{34}S values can also provide information on dietary protein intake, but their importance is certainly not as well understood.[15] It has been reported that δ^{34}S values may be correlated with δ^{15}N values and therefore may also track protein consumption.[16]

The mammalian hair cycle consists of four main stages – anagen (growing), catagen (regression), telogen (resting) and exogen (shedding) – with varying length for each phase.[17,18] The timing of the new hair cycle can vary as old hair may be shed immediately or stay intact for some time. Importantly, bears have two main hair types: guard hair and underfur. Guard hair is continuously grown throughout the spring to fall months, whereas underfur is suggested to grow over a shorter time-frame during the late summer to fall.[19,20] Molting of hair occurs once a year during the summer[21] followed by the cyclic production of new hair. Temporal information can be obtained by growth section analysis of whole and guard hair samples reflecting the time of tissue formation,[7] and underfur

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potentially reflecting autumn dietary intake.[19] However, synchronized activity of hair growth is not generally seen across the entire body or for various types of hair,[22] and much is still to be understood for any animal model.

Isotopic studies on bears have sampled hair from either single[23] or multiple sites,[7,11,24] or even without the exact anatomical site being specified.[12,21,25] Hair collected non-invasively such as by hair traps[26-28] and rub trees[29] results in sampling from unknown anatomical sites from the bear. Mammalian hair can show spatial variation across the body for certain species[30]; however, the current understanding of isotopic variation across anatomical sites for grizzly bears is limited.[20] Thus, a question remains as to whether sampling of hair from multiple locations is necessary for isotopic studies of grizzly bears. This study aimed to understand whether the isotopic ratios of carbon, nitrogen and sulfur differ between paired hair samples taken from the rump and shoulder of individual grizzly bears (*Ursus arctos*).

2 | METHODS

2.1 | Study area

Bears were sampled in the northern (69.55458°N to 68.55098°N;139.9921°W to 137.0783°W) and the southern (60.11084°N to 60.75001°N;133.8967°W to135.9416°W) areas of the Yukon, Canada. Geographically the bulk of the Yukon is a subarctic plateau interspersed by mountains. The major exception is the Arctic Coastal Plain, a narrower eastward continuation of the same region in Alaska, which slopes down to the Beaufort Sea from the British Mountains inland. Yukon has a sub-arctic continental climate with temperatures ranging between 36°C in the summer and −60°C in the winter. Long hours of daylight during the summer promote rapid growth which compensates to some extent for the cooler summer temperatures experienced north of 60° latitude. Except for the Arctic Coastal Plain and the higher mountains, most of Yukon lies below the treeline. Boreal forests cover 57% of the territory and the vegetation is classified as sub-arctic and alpine. The average annual precipitation ranges from less than 20 cm west of Whitehorse to more than 40 cm in Watson Lake.[31]

2.2 | Sample collection

Hair samples were collected from live-captured bears that were chemically immobilized by darting. A total of 81 (31 males; 50 females) grizzly bears (*Ursus arctos*) were captured in the Yukon territory, Canada (Figure 1), from 2004 to 2011 during the spring (March to July; $n = 75$) and fall (October; $n = 6$) seasons. The collected hair samples consisted of a mixture of underfur and guard hair which was cut from the rump and shoulder of each individual bear.

2.3 | Sample preparation

Samples of bear hair were cleaned of surface oils using repeated rinses of a 2:1 chloroform/methanol solution and then dried in an oven at +60°C overnight. Hair in the amount from 0.2 to 0.4 mg was cut and placed into tin cups, and 2 to 5 mg of vanadium pentoxide was added to promote sulfur combustion. The amount of sample or standard analyzed in the procedure was based on the criterion that the total amount of N or S in it must be within the range of 0.03 to 0.2 mg.

2.4 | Isotopic analyses

Hair samples were analyzed at Aquatech Enviroscience Laboratories Inc. (Victoria, BC, Canada) using a ECS 4010 thermal combustion elemental analyzer (Costech Analytical Instruments Inc., Valencia, CA, USA) coupled via a ConFlo III flow reducing interface to a

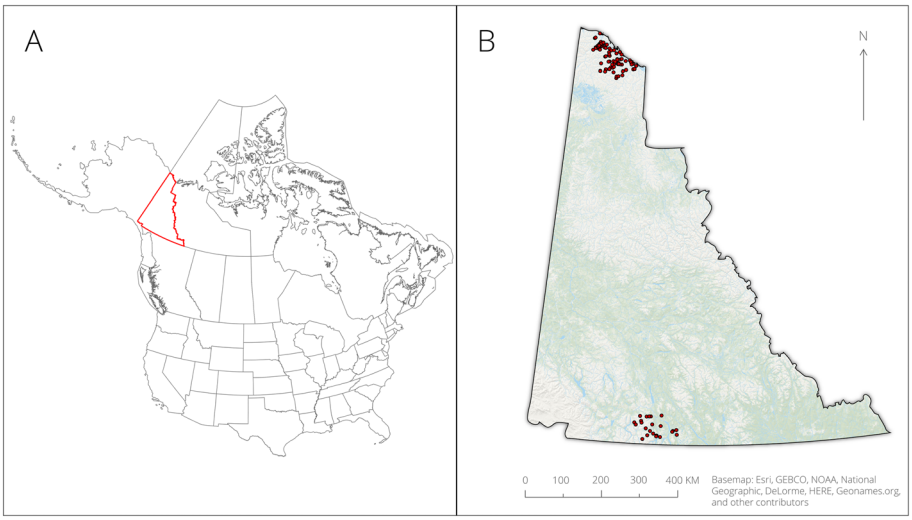

FIGURE 1 A, Map of North America outlining the Yukon territory in Canada. B, Map of sampling locations within the Yukon territory [Color figure can be viewed at wileyonlinelibrary.com]

DeltaPLUS Advantage continuous flow isotope ratio mass spectrometer (both from Thermo Fisher Scientific, Bremen, Germany). Individual samples were run as duplicates or triplicates. As a part of the laboratory quality assurance and quality control procedure, two types of secondary reference material in duplicates were analyzed before and after each batch of samples: dorm-2 (dogfish muscle reference material from National Research Council, Ottawa, ON, Canada) with δ^{13}C and δ^{15}N values of −17.17‰ and +14.33‰, respectively, and caffeine with δ^{13}C and δ^{15}N values of −42.02‰ and −0.95‰, respectively. Bovine liver with a δ^{34}S value of +7.50‰ was used as a reference material for δ^{34}S measurements. These 'in-house' standards were also used to monitor the long-term performance of the mass spectrometer. Each batch of samples also included a sample duplicate and a procedural blank. The following international reference standards (National Institute of Standards and Technology, Gaithersburg, MD, USA) were used for the scale normalization of the instrument: IAEA-N1 (ammonium sulfate reference material certified by the International Atomic Energy Agency, Vienna, Austria) with a δ^{15}N value of +0.43‰, IAEA-N2 with a δ^{15}N value of +20.32‰, Sucrose ANU with a δ^{13}C value of −10.43‰, NBS-22 OIL with a δ^{13}C value of −29.74‰, IAEA-S-1 (Ag2S) with a δ^{34}S value of −0.3‰ and NBS-127 (BaSO4) with a δ^{34}S value of −21.1‰.

2.5 | Instrumental precision

The instrumental precisions were 0.33‰, 0.25‰ and 0.40‰ for δ^{13}C, δ^{15}N and δ^{34}S values, respectively. The nitrogen, carbon and sulfur isotope data are reported in conventional δ-notation in units of per mil (‰) with reference to atmospheric nitrogen (air), Vienna Pee Dee Belemnite carbonate and Vienna Canon Diablo Troilite standards, respectively, and calculated according to the equation:

$$\delta^{i}\mathrm{E}_{\mathrm{std}} = \frac{R^{i}{}_{{}^{j}\mathrm{E}_{\mathrm{P}}}^{\mathrm{E}}}{R^{i}\mathrm{E}_{{}^{j}\mathrm{E}_{\mathrm{std}}}} - 1$$

where R = ratio of the abundance of heavier ^{i}E to lighter ^{j}E isotopes in the analyte substance P and std = standards.[32]

2.6 | Statistical analyses

The mean and standard deviation (σ) of pooled differences for the δ^{13}C, δ^{15}N and δ^{34}S values for all paired samples were calculated by taking the isotopic ratios of rump hair and subtracting the shoulder hair ratios. Bland–Altman (B&A) plot analysis was utilized for data visualization rather than a correlation graph as the purpose was to evaluate the agreement between the intervals rather than the relationship between the two.[33] B&A plots require the specification of the acceptable maximum difference limits *a priori*, and thus were specified with the instrumental error. This error was calculated according to the rule of error propagation,[34] $Q^2 = \sum_{i=1}^{p} x_i^2$, where x denotes individual measurements with an error . As B&A plots do not allow for the quantitative interpretation of reliability, paired t-tests were simultaneously conducted at the α = 0.05 level. Further analyses were conducted by separating data by sex. The generalized extreme studentized deviate (ESD) test[35] was used for outlier detection. All statistical analyses were run on R version 3.2.3.[36]

3 | RESULTS

The mean paired differences for pooled data (n = 81) were −0.04 ± 0.25‰ for δ^{13}C values, −0.13 ± 0.36‰ for δ^{15}N values and + 0.13 ± 1.67‰ for δ^{34}S values (Figure 2). Paired hair samples were not statistically significantly different for the pooled δ^{13}C and δ^{34}S data. The statistical results for the δ^{15}N data were significantly different at the α = 0.05 level; however, the mean paired difference fell within the instrumental error (Table 1). One outlier was detected with the ESD test for δ^{13}C values, three outliers for δ^{15}N values and five outliers for δ^{34}S values. Removal of the outliers did not alter the results of the statistical tests.

When data were separated by sex, male grizzly bears showed paired differences (n = 31) of −0.03 ± 0.30‰ for δ^{13}C values, −0.15 ± 0.30‰ for δ^{15}N values and −0.32 ± 1.63‰ for δ^{34}S values (Table 1). Females showed mean paired differences (n = 50) of −0.05 ± 0.21‰ for δ^{13}C values, −0.11 ± 0.39‰ for δ^{15}N values and +0.41 ± 1.68‰ for δ^{34}S values. The results from the paired t statistics showed no statistical significance between the rump and shoulder hair of both male and female grizzly bears with the exception of the δ^{15}N values. The differences were statistically significant but fell within the instrumental error, rendering the differences not meaningful. Furthermore, the time of year of sampling did not influence the results for either males or females.

Additional samples were retrieved from four grizzly bears (YT006, YT007, YT009 and YT029) at different dates over an extended period of time. This data was analyzed separately from the main pool of data to determine whether multiple sampling of individual bears influenced paired isotopic differences between hair from the rump and shoulder. The four bears were all females and they were sampled three times during the spring season from April to June of 2004 to 2008. The mean of the paired isotopic differences between the rump and shoulder hair sampled from individual bears ranged from −0.16 to +0.04‰ for δ^{13}C values, −0.42 to +0.38‰ for δ^{15}N values and −0.38 to +2.13‰ for δ^{34}S values (Figure 3).

4 | DISCUSSION

The results from this study indicate that the δ^{13}C, δ^{15}N and δ^{34}S values in mixed samples of guard hair and underfur do not show spatial variation between the rump and shoulder of grizzly bears. The δ^{15}N values in paired hair samples showed statistically significant differences, but this difference fell within the propagated instrumental error of 0.47‰ for δ^{15}N values. Thus, instrumental error captures the observed statistical offset, rendering the minor

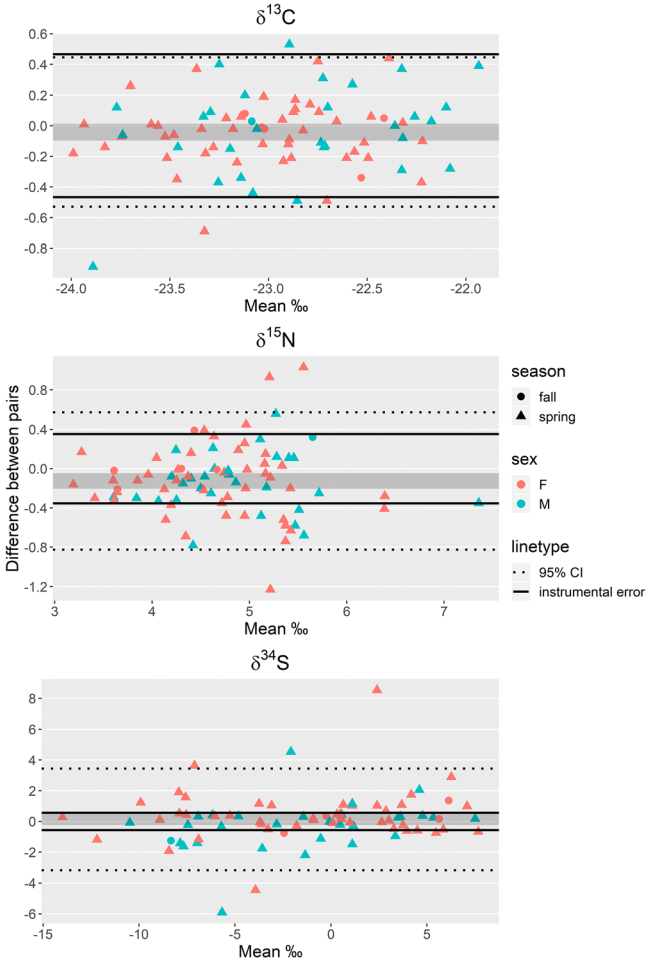

FIGURE 2 Bland–Altman (B&A) plot of the paired differences of $\delta^{13}C$, $\delta^{15}N$ and $\delta^{34}S$ values between hair from the rump and shoulder of all grizzly bears ($n = 81$). B&A plots require the specification of the acceptable maximum difference limits *a priori* and were set with the propagated instrumental errors for each isotope. 95% confidence intervals for the paired differences are shown by the dotted line. The shaded areas show the 95% confidence intervals for the mean differences and fell within the range of propagated instrumental errors for $\delta^{13}C$, $\delta^{15}N$ and $\delta^{34}S$ values of 0.47‰, 0.35‰ and 0.57‰, respectively [Color figure can be viewed at wileyonlinelibrary.com]

difference not meaningful. Our results are comparable with an analysis of eight paired hair samples taken from the back and rump by Mowat and Heard,[20] where very little spatial variation between the two sample locations was observed. Their results also utilized samples composed of a mixture of guard hair and underfur. In addition, mean paired differences of hair from the four bears that were sampled multiple times were within the instrumental precision for $\delta^{13}C$ values although larger variances were observed for the $\delta^{15}N$ and $\delta^{34}S$ values.

Paired differences in $\delta^{34}S$ values between the rump and shoulder hair exhibited larger variances than in the $\delta^{13}C$ and $\delta^{15}N$ values, although the overall differences of values between the rump and shoulder for both sexes were non-significant. Of the pooled data, a total of 8 pairs showed a difference in $\delta^{34}S$ values greater than 2‰. This may, in part, be explained by the high variability in the $\delta^{34}S$ values of the diet, since there can be considerable variability in the $\delta^{34}S$ values of local plants, both inter- and intra-plant, where $\delta^{34}S$ values can vary by up to 10‰ within a single plant.[37] In addition, terrestrial food sources can show a greater variance in their $\delta^{34}S$ values than in their $\delta^{13}C$ values, whereas marine food sources might reflect limited variability in their $\delta^{34}S$ values.[15] This difference might also be explained by the varying amounts of guard hair to underfur that were sampled, which is not known for this study.

Future research might examine guard hair and underfur separately to deal better with the issue of temporal variability between guard hair and underfur. It is known that underfur grows later in the summer season than guard hairs[16] and thus mixing should average out any isotopic differences in diet during the two growth phases. This may explain the homogeneity of isotopic signatures observed across the two anatomical sites examined in this study.

The absence of isotopic spatial variation across the two anatomical areas of grizzly bears demonstrates that hair sampling may be simplified to just one anatomical location, which is important for sampling of bears that are chemically immobilized. This method inevitably puts stress on the bear and thus it is best to minimize the time taken for the sampling procedure.[38] It further has important practical implications for non-invasive hair sampling, where common methods such as the use of hair traps result in hair being collected from various regions of the body and are often not site-specific.[26]

TABLE 1 Statistical results for paired t tests on rump and shoulder hair of grizzly bears for $\delta^{13}C$, $\delta^{15}N$ and $\delta^{34}S$ values

	$\delta^{13}C$		$\delta^{15}N$		$\delta^{34}S$	
	Pooled		Pooled		Pooled	
t	−1.505		−3.172		0.706	
df	80		80		80	
p	0.136		0.002		0.482	
Mean of differences	−0.04		−0.13		+0.13	
	Male	Female	Male	Female	Male	Female
t	−0.508	−1.687	−2.686	−2.067	−1.101	−1.749
df	30	49	30	49	30	49
p	0.615	0.098	0.012	0.044	0.482	0.087
Mean of differences	−0.03	−0.05	−0.15	−0.11	−0.32	+0.41

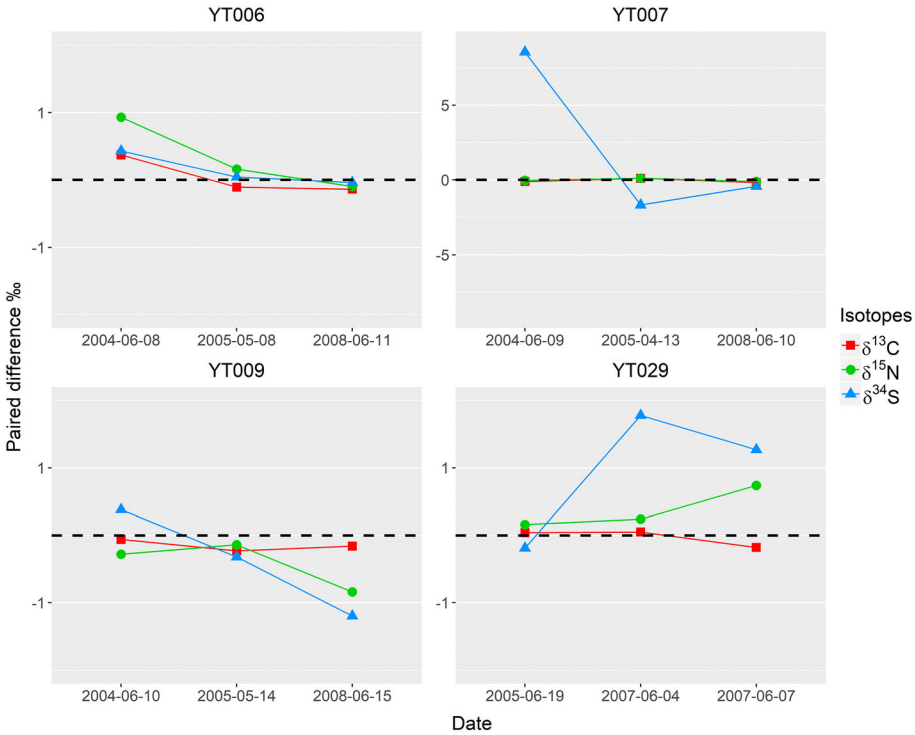

FIGURE 3 Paired differences of $\delta^{13}C$, $\delta^{15}N$ and $\delta^{34}S$ values between hair from the rump and shoulder of individual bears sampled multiple times over an extended time-period. It is evident that the sampling at differing times a year did not alter the relationship between rump and shoulder. The connection of data points is used here, in part, because there is a temporal relationship of three data points to a single bear, and to clearly differentiate one bear from another [Color figure can be viewed at wileyonlinelibrary.com]

5 | CONCLUSIONS

The paired comparison of $\delta^{13}C$, $\delta^{15}N$ and $\delta^{34}S$ values in hair sampled from the rump and shoulder of grizzly bears showed no meaningful differences. The main finding of this study was that it is indeed not necessary to sample hair from both the rump and the shoulder for isotopic studies of grizzly bears for hair consisting of a mixture of guard hair and underfur. Setting these results more widely, it may be that other animals being sampled in this way will not need to be sampled from multiple anatomical locations.

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